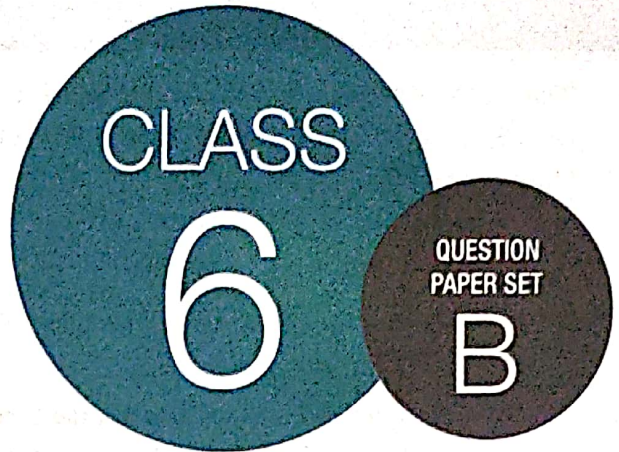


**SOF INTERNATIONAL
MATHEMATICS OLYMPIAD
2022-23**



DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Total Questions: 50 | Time: 1 hr.

Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Roll No.** and **Mobile Number** clearly on the **OMR Sheet** and do not forget to sign it. We will share your marks / result and other information related to SOF exams on your mobile number.
3. The Question Paper comprises four sections:

Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles on the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.
Q.16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.
A. 11.450 kg B. 11.000 kg C. 11.350 kg D. 11.250 kg
As the correct answer is option A, you must darken the circle corresponding to option A on the OMR Sheet.
7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in the space provided on this page before attempting the paper.

16. ● (B) (C) (D)



Name:.....

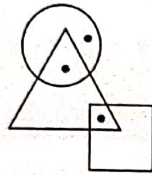
SOF Olympiad Roll No.:..... Contact No.:.....

LOGICAL REASONING

1. Pointing towards Ankit, a lady said, "His mother is the only daughter of my mother." How is that lady related to Ankit?

A. Mother-in-law
B. Daughter
C. Mother
D. Sister-in-law

2. Which of the following options satisfies the same conditions of placement of dots as in the given figure?



A.

B.

C.

D.

3. Which of the following Venn diagrams best represents the relationship amongst, "Judge, Police and Thief"?

A.

B.

C.

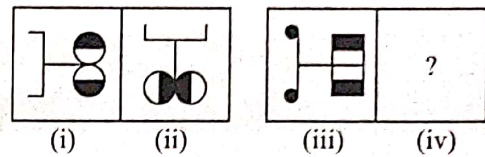
D.

4. Select a term from the options which will continue the given series.

B4M, D7L, F11K, H16J, ?

A. J22I
B. I20H
C. J20C
D. K19I

5. There is a certain relationship between figures (i) and (ii). Establish the similar relationship between figures (iii) and (iv) by selecting a suitable figure from the options that will replace the (?) in figure (iv).



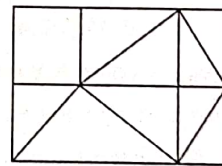
A.

B.

C.

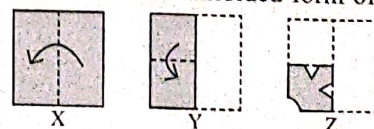
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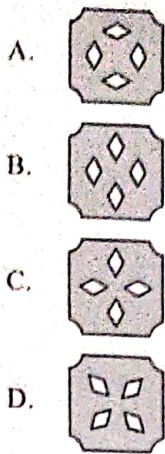
6. Find the number of triangles formed in the given figure.



A. 10
B. 12
C. 11
D. More than 12

7. The given question consists of a set of three figures X, Y and Z which shows the sequence of folding of a piece of paper. Fig. (Z) shows the manner in which the folded paper has been cut. Select a figure from the options which shows the unfolded form of Fig. (Z).

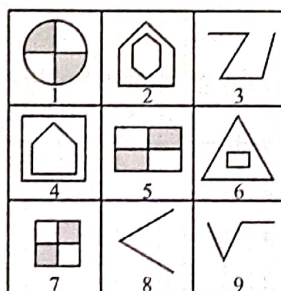




8. How many such symbols are there in the given arrangement each of which is immediately preceded and immediately followed by an even number?

5 6 # A 3 Z @ U 4 M 2 ? 6 • 8 T S © N

- A. None
B. One
C. Two
D. Three
9. Group the given figures into three classes on the basis of their identical properties using each figure only once.



- A. 1, 2, 4; 3, 5, 8; 6, 7, 9
B. 1, 5, 7; 2, 3, 9; 4, 6, 8
C. 1, 5, 7; 2, 4, 6; 3, 8, 9
D. 1, 3, 4; 2, 5, 7; 6, 8, 9
10. Some letters are given which are numbered 1, 2, 3, 4, 5 and 6. Find the combination of numbers from the options so that the letters are arranged accordingly to form a meaningful English word.

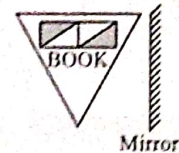
L J N U G E
1 2 3 4 5 6

- A. 2, 4, 3, 5, 1, 6
B. 2, 1, 3, 5, 4, 6
C. 3, 1, 4, 2, 6, 5
D. 4, 1, 2, 5, 3, 6

11. In a certain code language, if CREDIT is coded as CDEIRT, then how will SAFEST be coded in the same code language?

- A. TBGFTU
B. AEFSSST
C. ASEFTS
D. TSSFEA

12. Find the correct mirror image of the given figure.

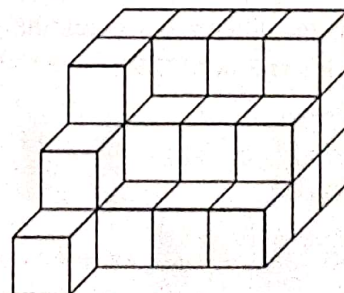


- A.
- B.
- C.
- D.

13. If '+' stands for division, '÷' stands for multiplication, '×' stands for subtraction and '-' stands for addition, then find the value of $14 \div 6 - 8 + 4 \times 10$.

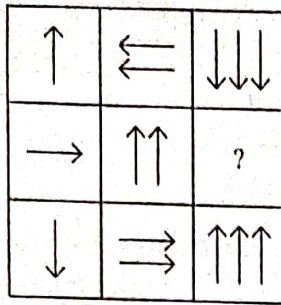
- A. 70
B. 150
C. 76
D. 86

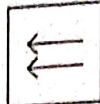
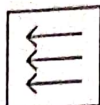
14. Count the number of cubes in the given figure.



- A. 20
B. 27
C. 25
D. 24

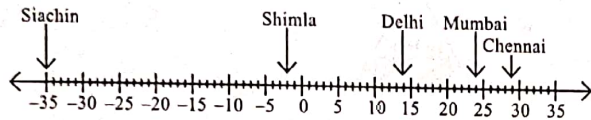
15. Select a figure from the options which will complete the given figure matrix.



- A. 
- B. 
- C. 
- D. 

MATHEMATICAL REASONING

16. Given number line represents the average temperature (in °C) of different places in January. Find the temperature difference between Siachin and Chennai.



- A. 6°C
B. 65°C
C. -6°C
D. None of these

17. What is the fraction of vowels in the word "MATHEMATICS"?

- A. $\frac{5}{11}$
B. $\frac{7}{11}$
C. $\frac{6}{11}$
D. $\frac{4}{11}$

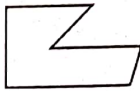
18. What is the difference between the place value and face value of 3 in 637258?

- A. 29997
B. 15980
C. 30003
D. None of these

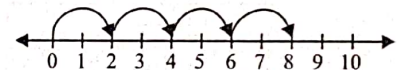
19. Find the number of prime factors of 7350.

- A. 2
B. 4
C. 3
D. 5

20. Which of the following is not a polygon?

- A. 
- B. 
- C. 
- D. 

21. What does the given number line represent?



- A. 2×8
B. 2×4
C. $8 \div 2$
D. $8 - 2$

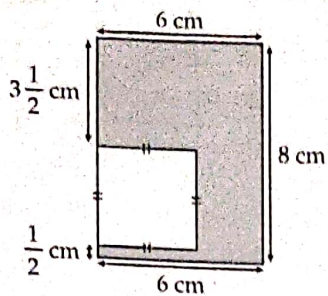
22. If $x : y = 4 : 5$, then $(4x + 5y) : (5x - 2y)$ is equal to _____.

- A. 41 : 10
B. 4 : 5
C. 16 : 25
D. 3 : 10

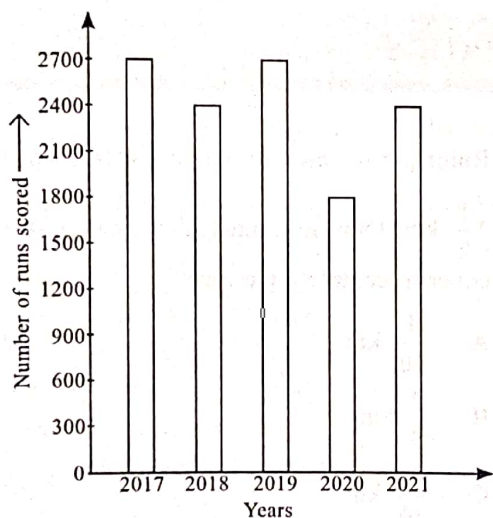
23. Vicky is 5 years older than his cousin. If age of Vicky is x years, then find the age of his cousin (in years).

- A. $x + 5$
B. $x \times 5$
C. $x \div 5$
D. $x - 5$

24. What is the area of the shaded part in the given figure?

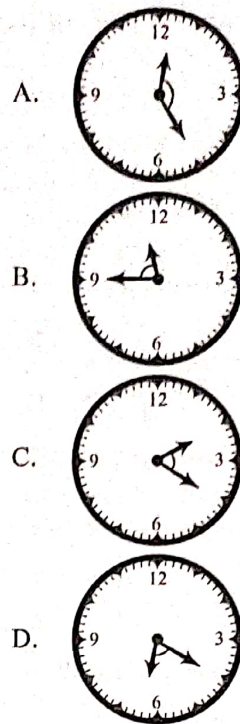


- A. 16 cm^2
 B. 24 cm^2
 C. 32 cm^2
 D. 48 cm^2
25. The value of $0.90 + 34.77 - 23.03$ is _____.
- A. 21.63
 B. 12.64
 C. 16.67
 D. 18.59
26. The given bar graph represents the total number of runs scored by a player in five different years across all formats of cricket. Study the graph carefully and answer the following question.



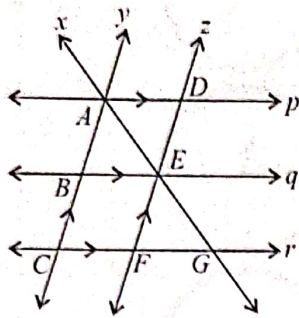
How many more runs were scored by the player in years 2018, 2019 and 2021 altogether than in years 2017 and 2020 together?

- A. 8350
 B. 7350
 C. 9350
 D. 3000
27. In which of the following clocks, obtuse angle will be formed between the hour hand and the minute hand of the clock?



28. The decimal form of $75 \frac{491}{1000}$ is
- A. 7504.91
 B. 7500.491
 C. 75.491
 D. 75.0491
29. Which of the following has exactly two lines of symmetry?
- A. Equilateral triangle
 B. Isosceles triangle
 C. Rectangle
 D. Square
30. What is the numeral form of the number, "Two hundred fifty three million five hundred three thousand four hundred two" in International System of Numeration?
- A. 253,530,402
 B. 253,503,402
 C. 253,503,420
 D. 253,053,402
31. If  +  +  = 400 and  +  = 240, then find the value of  and  respectively.
- A. 160, 80
 B. 140, 100
 C. 80, 160
 D. 100, 140

32. In the given figure, the number of pairs of parallel lines is _____.

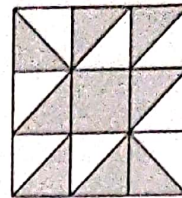


- A. 4
B. 3
C. 2
D. 5

33. Which of the following options is correct?

- A. Smallest natural number is 0.
B. Every whole number has a successor.
C. Predecessor of a two digit number is always a two digit number.
D. Smallest whole number is 1.

34. Find the shaded fraction of the given figure.



- A. $\frac{5}{9}$
B. $\frac{1}{3}$
C. $\frac{1}{9}$
D. $\frac{2}{9}$

35. The given data shows the number of goals scored by a player in 24 different football matches. Study the data carefully and answer the following question.

2, 3, 2, 1, 0, 4, 3, 1, 2, 0, 3, 2, 1, 3, 0, 2, 5, 2, 4, 3, 1, 2, 1, 3

How many times the player scored at least 2 goals?

- A. 13
B. 14
C. 15
D. 16

EVERYDAY MATHEMATICS

36. Arpita takes 5 rounds of a rectangular park length of which is 7 m less than two times of its breadth. If breadth of the rectangular park is 65 m, then find the total distance travelled by Arpita.

- A. 1680 m
B. 1560 m
C. 1880 m
D. 1660 m

37. The students in a class can be divided into groups of 2, 3, 5 and 6. What is the least number of students this class can have?

- A. 40
B. 30
C. 35
D. 42

38. The cost of 5 eggs is ₹ 27.50 and cost of 5 omelettes is ₹ 60. Find the ratio of cost of one egg to the cost of one omelette.

- A. 12:25
B. 25:12
C. 24:11
D. 11:24

39. Rohit participated in a race of $1\frac{3}{5}$ km. If he runs

$1\frac{1}{2}$ km, then how much distance he still needs to cover to complete the race?

- A. $\frac{1}{10}$ km
B. $\frac{2}{5}$ km
C. $\frac{3}{10}$ km
D. $\frac{3}{5}$ km

40. The cost of cultivating a square field at ₹ 17 per sq. m is ₹ 612. Find the side of the field.

- A. 9 m
B. 8 m
C. 6 m
D. 10 m

41. Rohan saved ₹ 2860 from his pocket money in a certain year. What is the Roman number for this saving amount?

- A. MMDCCCLX
B. MDCCCLX
C. MMDCCCLX
D. MMDCCCLIX

- B. 4,38,114
C. 5,48,224
D. 5,38,114

42. Arun has ₹ 125 in his savings account. He withdraws ₹ 117, makes a deposit of ₹ 45 and then withdraws another ₹ 69. Find the amount left in his account. (Write the amount as an integer.)

- A. ₹ (-16)
B. ₹ 16
C. ₹ 30
D. ₹ (-30)

43. A company X produces two types of laptops *i.e.*, P and Q . Total production of both types of laptops in a certain year is 8,75,677. If total production of P type laptops is 3,37,563, then find the total production of Q type laptops.

- A. 4,48,224

44. Prerna had ₹ 5000. She bought 2 boxes of chocolates for ₹ 52.98 each, 3 packets of chips for ₹ 12.98 each and 2 books for ₹ 300.05 each. How much money is left with her?

- A. ₹ 4255
B. ₹ 4750
C. ₹ 4000
D. ₹ 3875

45. Kanika's teacher asked her to round off 1248, 3127 and 4105 to the nearest tens. After rounding off the numbers, the estimated sum will be _____.

- A. 7500
B. 8000
C. 8480
D. None of these

ACHIEVERS SECTION

46. Match the following and select the correct option.

Column-I

Column-II

- (P) The sum of place values of the digits 5 and 8 in the number 95807 is _____.
(Q) 8 hundreds more than smallest 4-digit even number is _____.
(R) Greatest 4-digit number having all different digits such that 3 is at tens place is _____.

- | | (P) | (Q) | (R) |
|----|-------|-------|-------|
| A. | (iii) | (i) | (ii) |
| B. | (ii) | (i) | (iii) |
| C. | (iii) | (ii) | (i) |
| D. | (i) | (iii) | (ii) |

48. Fill in the blanks and select the correct option.

- (i) The value of $57.295 + 108.280 - 33.912$ is (P).
(ii) $50 + 5 + \frac{5}{10} + \frac{5}{100}$ in decimal form can be written as (Q).
(iii) Rashmi purchased a chocolate for ₹ 12.50, a pastry for ₹ 5.75 and a packet of chips for ₹ 8.38. Total amount of money spent by her is (R).

- | | (P) | (Q) | (R) |
|----|---------|--------|---------|
| A. | 131.663 | 505.55 | ₹ 20.15 |
| B. | 130.405 | 55.55 | ₹ 28.92 |
| C. | 131.663 | 55.55 | ₹ 26.63 |
| D. | 128.35 | 50.55 | ₹ 20.50 |

47. Read the given statements carefully and select the correct option.

Statement-I: One-fourth of a revolution is a right angle.

Statement-II: A person is facing towards North. He turns clockwise to face towards South-West. So, he turned $\frac{5}{8}$ of a revolution clockwise.

- A. Statement-I is true but Statement-II is false.
B. Statement-I is false but Statement-II is true.
C. Both Statement-I and Statement-II are true.
D. Both Statement-I and Statement-II are false.

49. Answer the following questions and select the correct option.

- (a) A path (outside) of width 50 cm is made around a rectangular field of dimensions 12 m and 8 m. Find the area of that path.
(b) A wire of length 68 cm is bent into the form of a square. Find the area of the square so formed.

- | (a) | (b) |
|---------------------|--------------------|
| A. 21 m^2 | 215 cm^2 |
| B. 17 m^2 | 200 cm^2 |
| C. 26 m^2 | 68 cm^2 |
| D. 21 m^2 | 289 cm^2 |

50. The given pictograph shows the number of laptops sold by five different shops in a month. Study the pictograph carefully and answer the following questions.

Shops	Number of laptops sold
P	   
Q	  
R	    
S	    
T	  
Each  represents 4 laptops.	

- (i) How many more laptops were sold by shop S than shop Q?
- (ii) Find the fraction of the number of laptops sold by shops P and R together to the total number of laptops sold by all the shops.

- (i) (ii)
- A. 12 $\frac{13}{27}$
- B. 10 $\frac{17}{38}$
- C. 8 $\frac{11}{14}$
- D. 10 $\frac{14}{15}$

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